

Ibrahim Abiodun Imam

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SUMMARY

Ph.D. candidate in Chemical Engineering specializing in computational chemistry, cheminformatics, molecular data science, and AI/ML for drug discovery. Experienced in molecular dynamics, docking, QM/MM, alchemical free-energy calculations, custom force-field parameterization, and machine-learning driven discovery. Industry experience at GSK supporting large-scale chemical data workflows, hit-calling analysis, medicinal chemistry decision-making, SQL/BigQuery retrieval pipelines, and ML design for drug-discovery.

EDUCATION

University of Kentucky, Lexington KY

Aug 2022 - Nov 2026 (Expected)

Ph.D. in Chemical Engineering (GPA: 3.92)

SKILLS & COMPETENCIES

- **Computational Chemistry and CADD:** Molecular dynamics (GROMACS, OpenMM, Amber), docking (Autodock Vina, MOE), QM/MM (Orca/Amber), alchemical calculations, FEP, MM/GBSA, enhanced sampling (PLUMED), protein-ligand binding analysis (MDAnalysis, Pymol, chimera).
- **Cheminformatics & Molecular Data Science:** RDKit, OpenBabel, DEL data analysis, hit calling, SAR analysis, physicochemical property analysis, molecular dataset curation (ChEMBL).
- **AI/ML for Drug Discovery:** Deep learning adaptation and PEFT (ESM2, ProteinMPNN, chemprop), ensemble learning, molecular property prediction, model evaluation, PyTorch, scikit-learn, XGBoost, LightGBM.
- **Programming, Data & Automation:** Python, SQL, Google BigQuery, NumPy, Pandas, Bash/Linux, Git, HPC, TIBCO Spotfire.

EXPERIENCE

Cheminformatics Co-op

Jan 2026 – June 2026

GSK plc, Cambridge MA

- Implemented ensemble learning strategy for target-specific hit identification, improving average precision by 25%.
- Evaluated hits generated from collation-based statistical hit-calling workflows, expanding the hit set by ~10%.
- Supported large-scale triage of over 500M compounds to about 1000 candidates for SOS1/2 target.
- Designed SQL workflows to retrieve previously calculated chemical properties, reducing property-calculation tasks from ~6 hours to under 1 minute.
- Developed TIBCO Spotfire dashboards for medicinal chemistry to enable rapid visualization of hit results and SAR-relevant trends.

Graduate Research Assistant

Aug 2022 - Present

University of Kentucky Lexington, KY

- Integrated MD, docking, and QM/MM calculations to investigate mutation-induced EGFR drug resistance and recommend 12 candidate compounds with potential to overcome resistance-associated mutations.
- Applied alchemical free-energy and FEP methods to quantify mutation-dependent effects on GTP/GDP binding in KRAS.
- Used well-tempered metadynamics to identify p53 breathing conformations associated with early structural collapse, highlighting conformational markers of destabilization.
- Combined protein language models with MD simulations to support the discovery of functional antibiofouling and antifreeze peptides.
- Automated HPC-based molecular simulation and analysis workflows using Python and Bash, improving reproducibility and reducing manual setup time.

Graduate Research Fellow

Jun 2023 - Aug 2023

LBNL, Berkeley CA

- Enhanced NWChemEx capabilities for biochemical system modeling and improved scalability on HPC resources.
- Built automation scripts for benchmarking and regression testing to validate software changes.
- Conducted benchmarking analyses and documented recommendations for maintainable workflows.

SELECTED PUBLICATIONS

- Imam I.A., et al. Discovering naturally occurring antifreeze peptides from microbiome by integrating protein language models and molecular dynamics simulation. *J. Mater Chem B* (2026).
- Imam, I. A., et al. L858R/L718Q and L858R/L792H mutations of EGFR inducing resistance against osimertinib. *Proteins: Structure, Function, and Bioinformatics* (2025).
- Imam I.A., et al. In-Silico Predictions of Drug Resistance in Lung Cancers with EGFR Mutation. *PASC '25: Proceedings of the Platform for Advanced Scientific Computing Conference* (2025).
- Imam, I. A., et al. Integrating protein language model and molecular dynamics simulations to discover anti-biofouling peptides. *Langmuir* (2024).

TRAINING & WORKSHOPS

- Virtual School on Machine Learning for Molecules (Apr 2025): data cleaning, molecular featurization, ML/DL
- MDAnalysis/MolSSI Workshop, Arizona State University (Jun 2024): scientific software development and version control.
- NSF i-CoMSE Methods for Enhanced Sampling (Apr 2023): umbrella sampling, metadynamics, replica exchange.